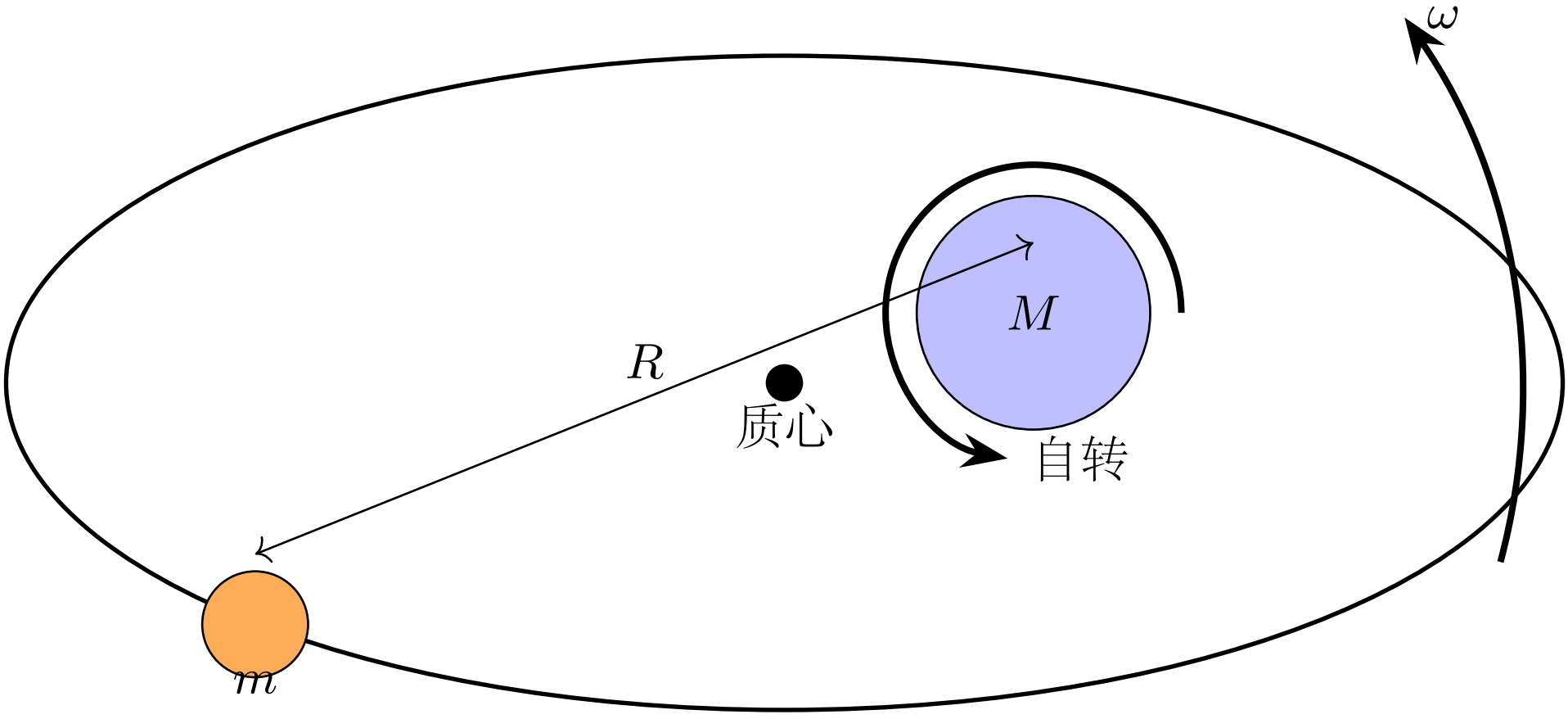
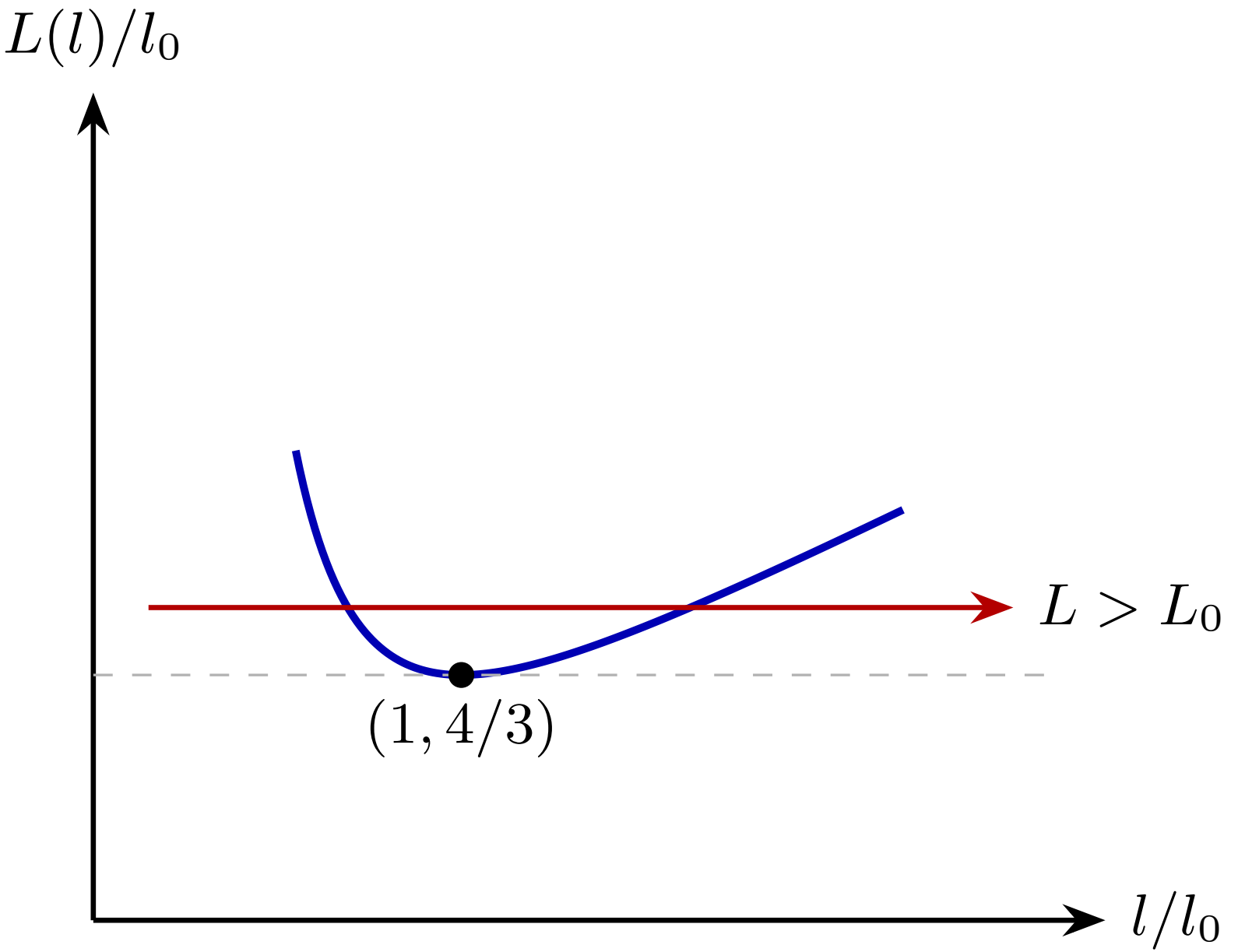
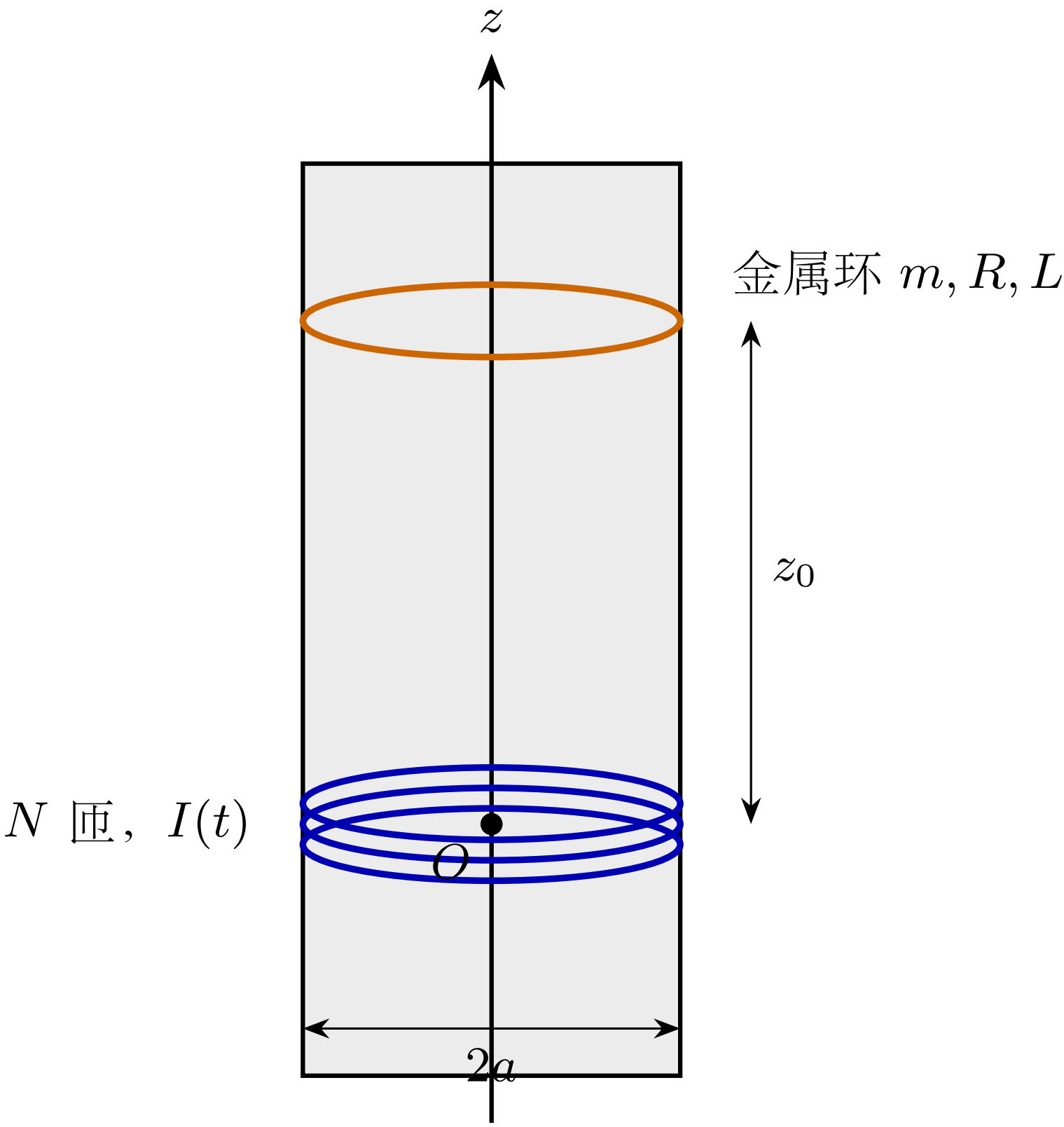


双星的公转与星体 M 的同向自转



横坐标刻度：图距 2 对应 $l/l_0 = 1$

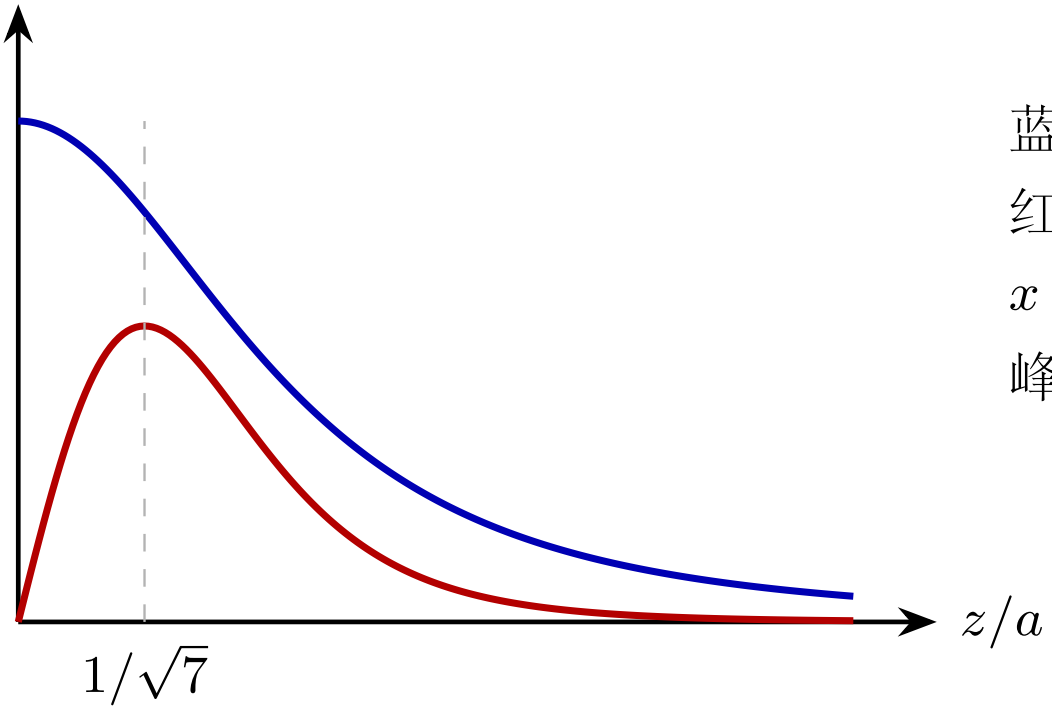
右支局部稳定；临界为驻点拐折，非稳定极小



线圈各匝集中在 $z = 0$，铁芯足够长

环半径同为 a；示意图不按横、纵同一比例

归一化曲线

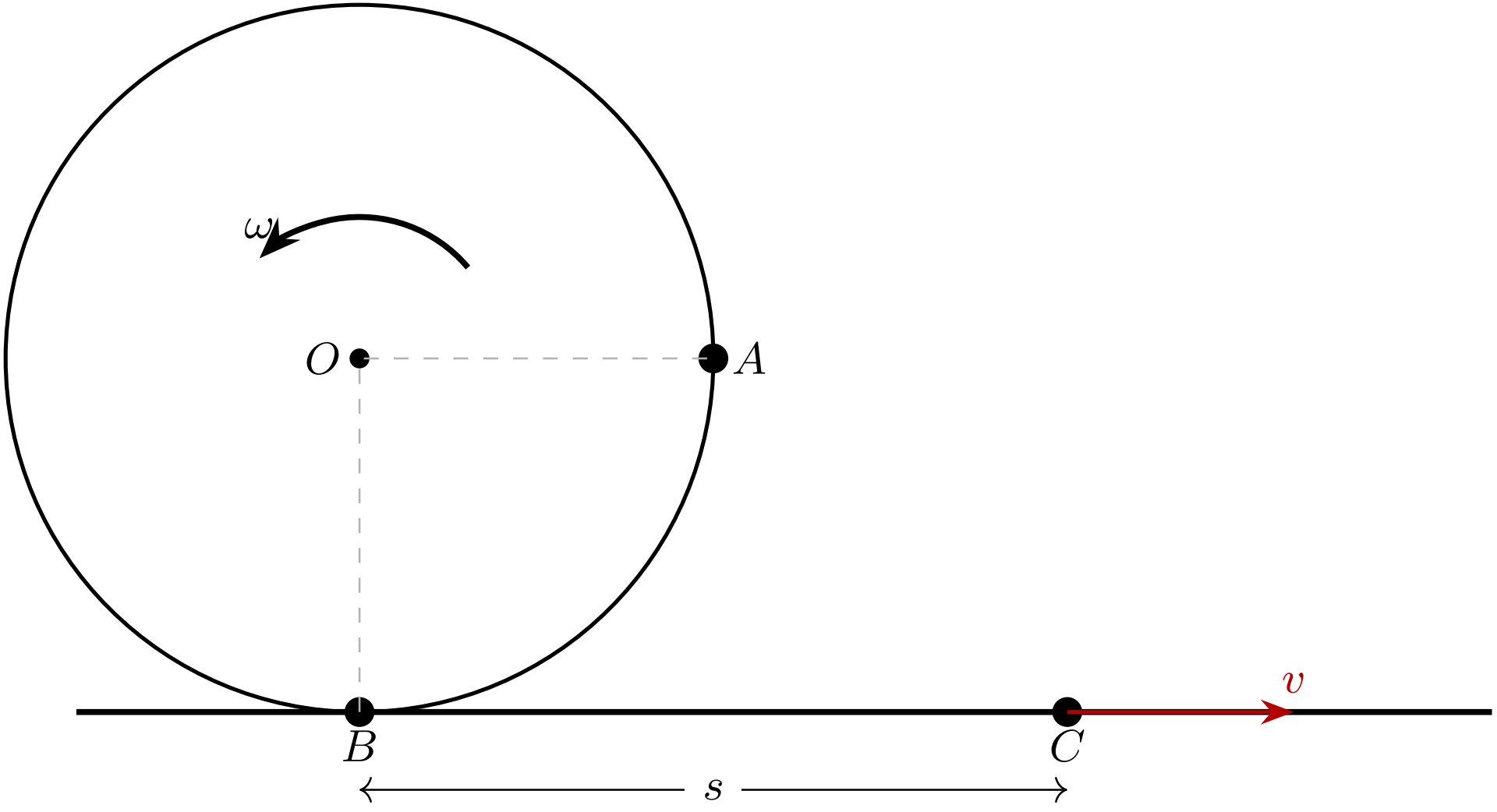


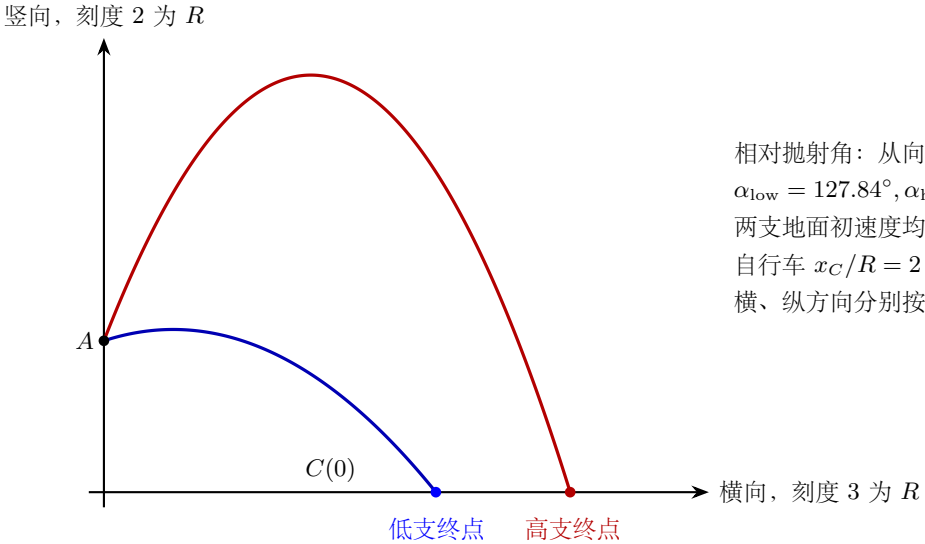
蓝：$M/M(0)$（纵向乘 3）

红：$x/(1+x^2)^4$（乘 8）

$x = z/a$

峰值右侧 $d\langle F\rangle/dz < 0$ 为稳定支


ω
O
A
B
C
v
s



相对抛射角：从向上顺时针量

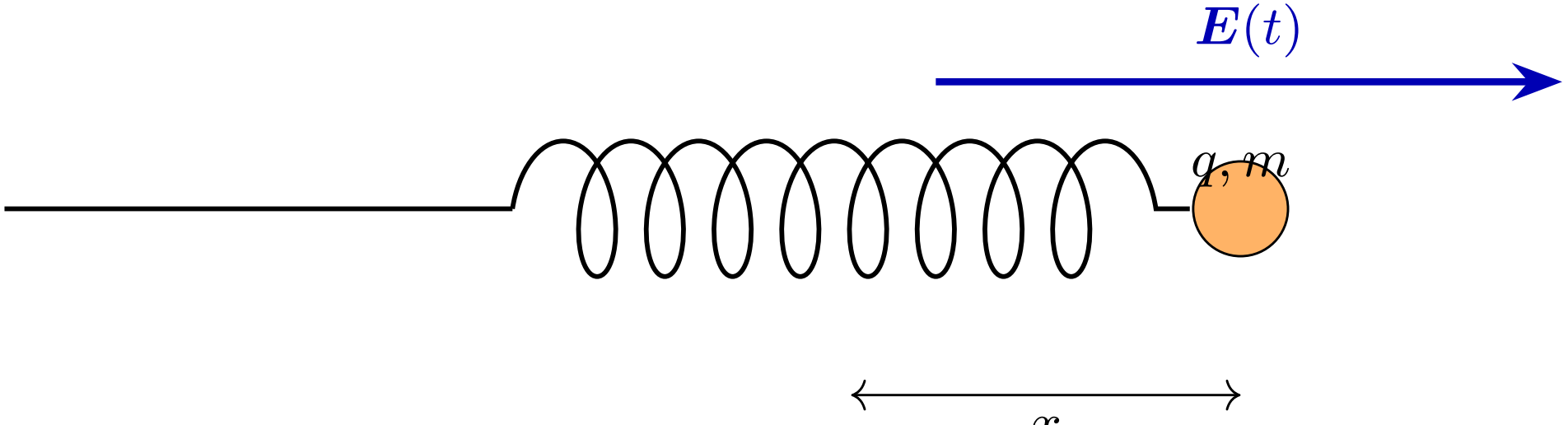
$\alpha_{\text{low}} = 127.84^\circ, \alpha_{\text{high}} = 29.14^\circ$

两支地面初速度均向上

自行车 $x_C/R = 2 + \tau/4$

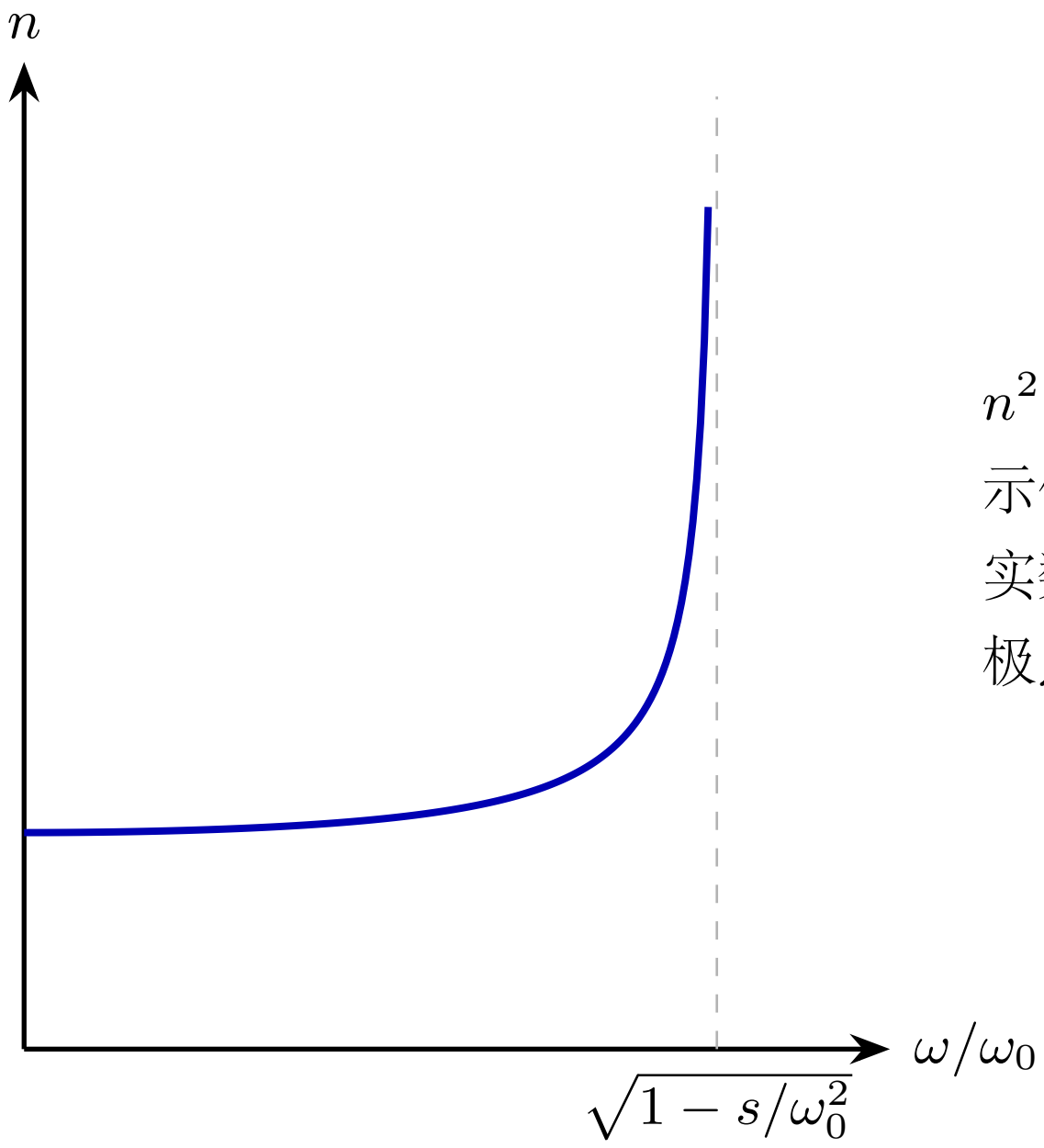
横、纵方向分别按注明刻度阅读

束缚电荷的受迫振动



先以 E_{ext} 求振幅 A

再求 $E_{\text{in}} = E_{\text{ext}} - bA$，不作自洽回代



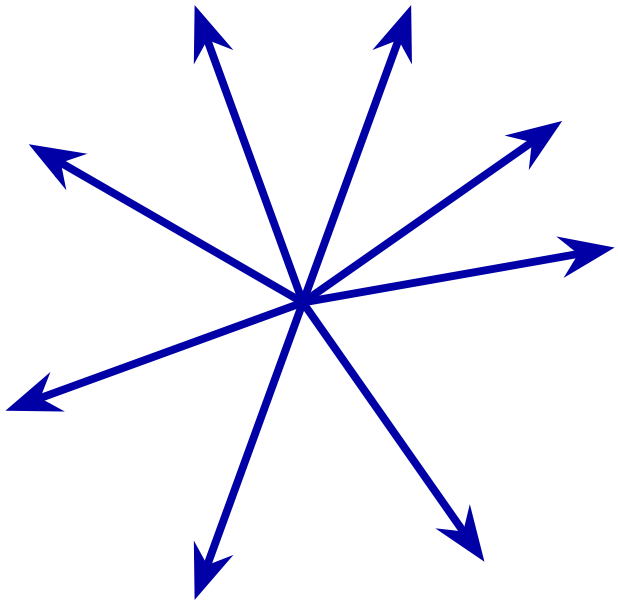
$n^2 = (\omega_0^2 - \omega^2)/(\omega_0^2 - s - \omega^2)$

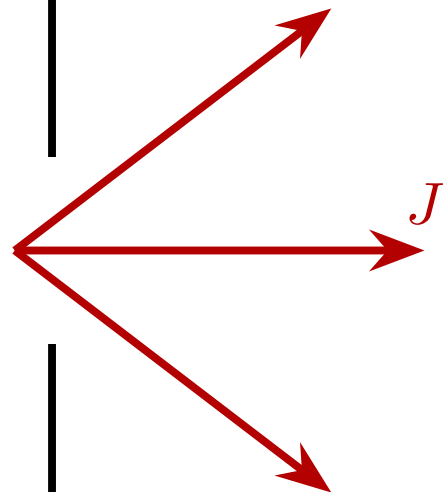
示例 $s/\omega_0^2 = 0.36$

实数低频支到场比极点为止

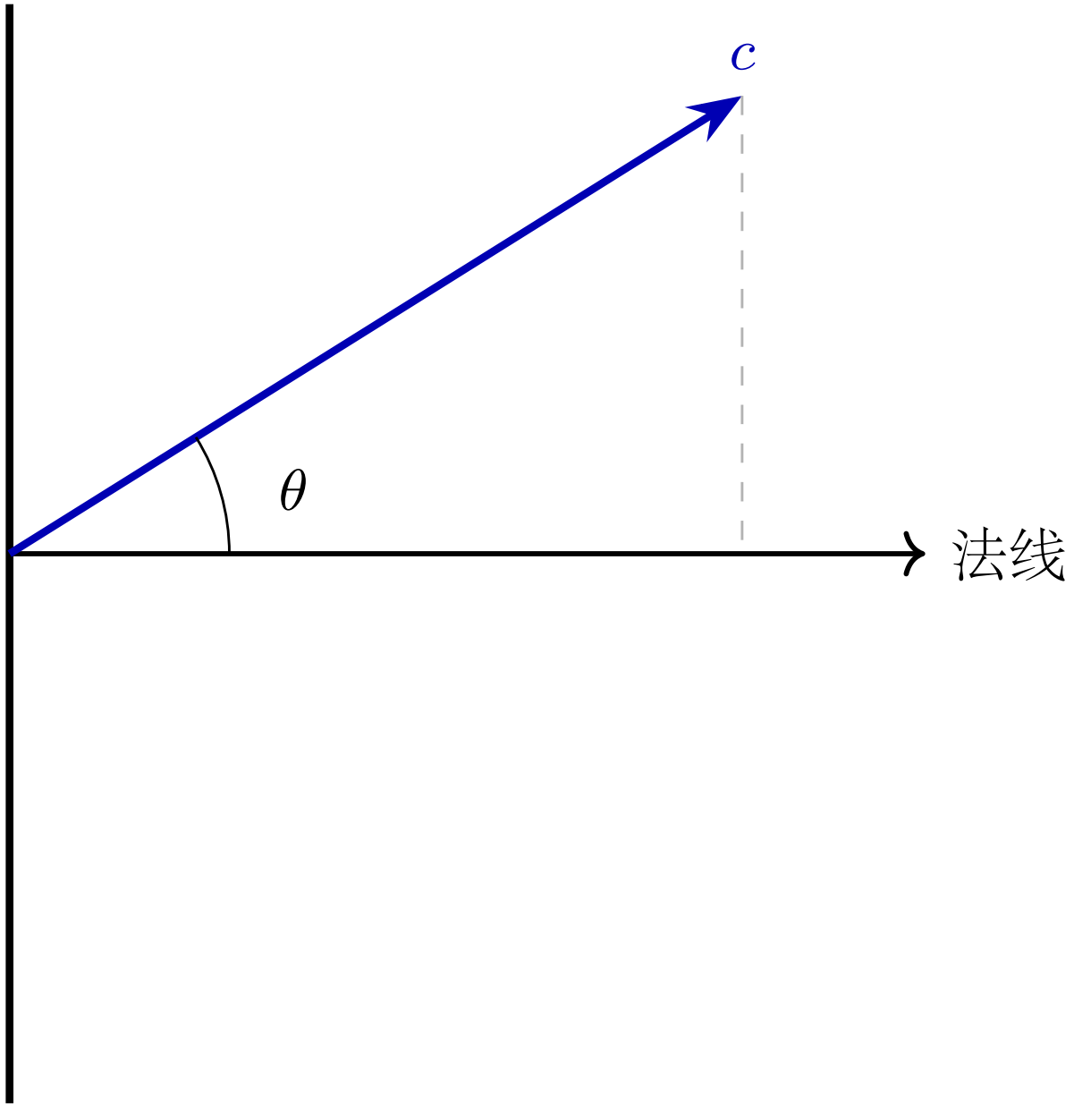
极点不在 $\omega = \omega_0$

热辐射空腔，能量密度 u





小孔

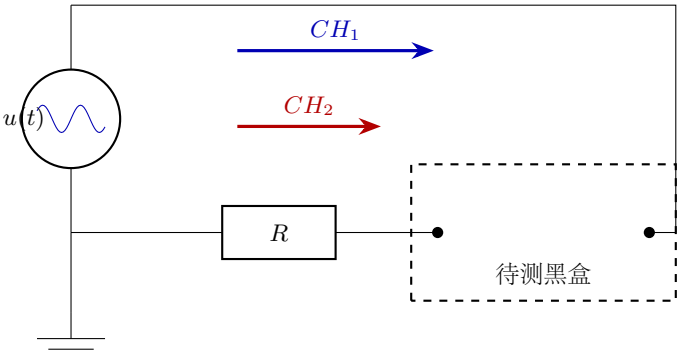


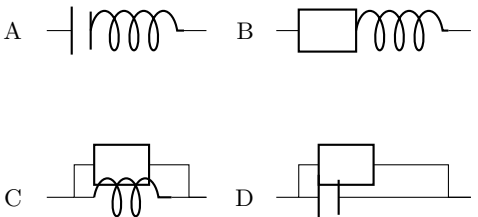
$$d\Omega = \sin\theta \, d\theta \, d\varphi$$

$$v_n = c\cos\theta$$

$$J = \frac{uc}{4}$$

$$P = \frac{u}{3}$$


CH_1
CH_2
$u(t)$
R
待测黑盒

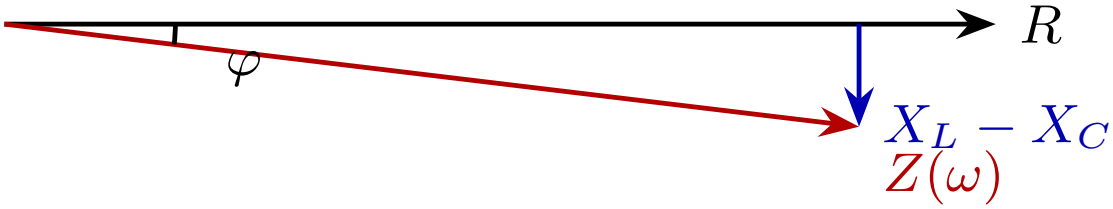

A
B
C
D

原值 $\varphi = -0.11938$ rad

$X_L - X_C = -3\omega L$

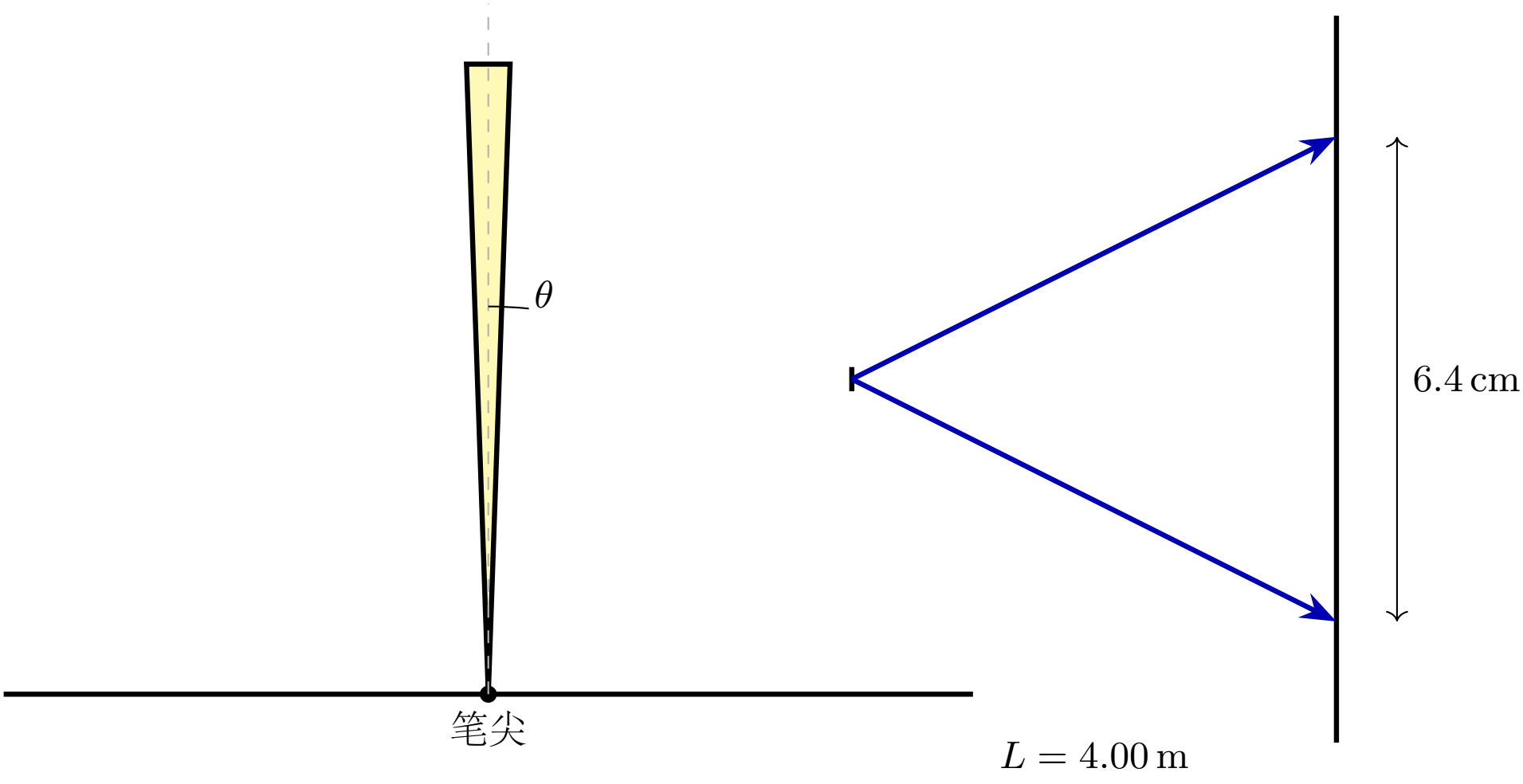
$\tan|\varphi| = 3\omega L/R$

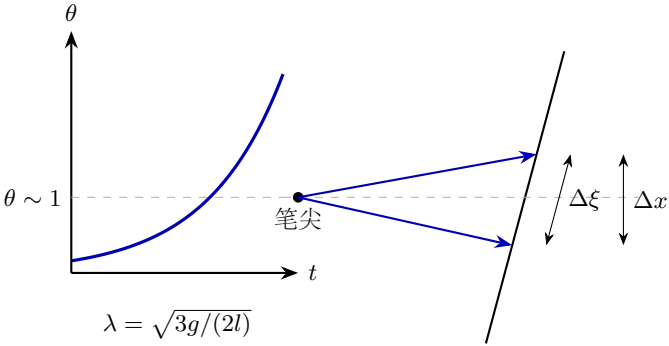
二倍频：$2\omega L = 1/(2\omega C)$



按 19 微秒真实相角绘制；三档无合规解

190 微秒只作校勘方案，另求 $L = 13.4\,\mathrm{mH}, C = 0.473\,\mu\mathrm{F}$


θ
笔尖
6.4 cm
L = 4.00 m



$\lambda = \sqrt{3g/(2l)}$

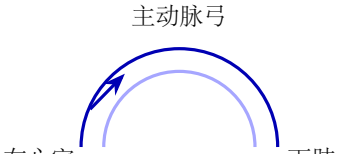
增长图仅用于线性近似

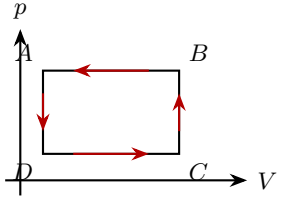
$\Delta x \simeq \Delta\xi \cos 15^\circ$

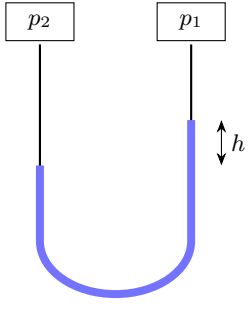
$d \simeq 25/\cos 15^\circ = 25.9\,\mu m$

图中角为 15 度；衍射角适当放大

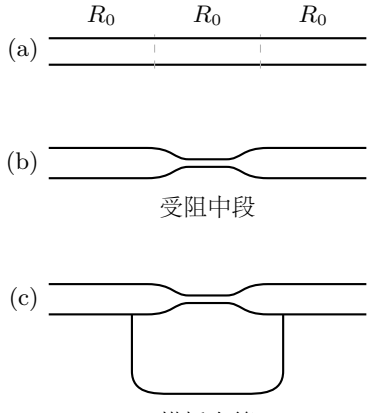
圆锥仅按局部宽度解释暗纹



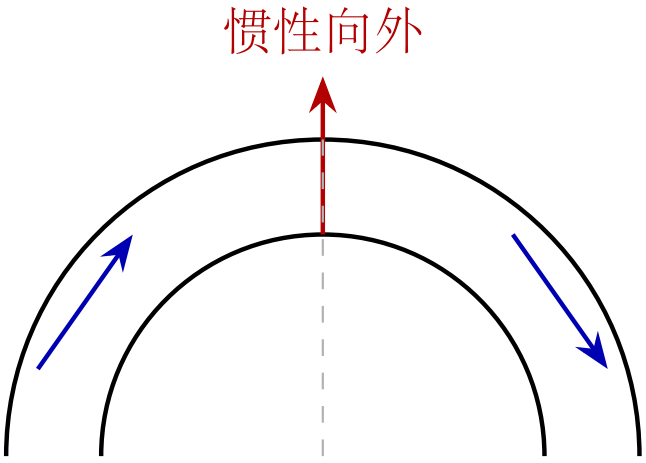




液柱测压装置



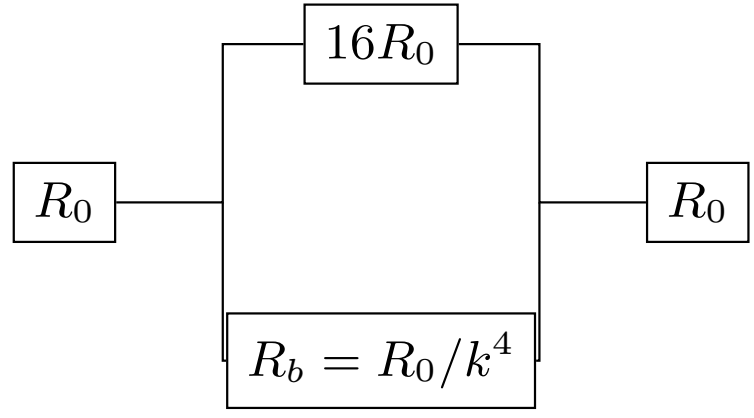
搭桥血管



局部：$dp/dr = \rho v^2/r$

$\Delta p_{\mathrm{loc}} = 9.1\,\mathrm{Pa}$

整体：$\bar{p}_{\mathrm{dyn}} = 2\rho Q v/A_{\mathrm{proj}} = 5.34\,\mathrm{Pa}$



$R_{\mathrm{eq}} = 2R_0 + (16R_0 \parallel R_b)$

$k = (15/16)^{1/4} = 0.984\ldots \simeq 0.98$

局部压差与整体平均压强对应不同面积，数值不能互换

$Q = 8.33 \times 10^{-5}\,\mathrm{m}^3/\mathrm{s}, v = 0.118\,\mathrm{m/s}$；膜张力 α 的单位为 $\mathrm{N/m}$